

PROJECT ER DIAGRAM

MULTIPLAYER VIDEO GAME DATA MANAGEMENT AND ANALYTICS SYSTEM

LAB GROUP: 5

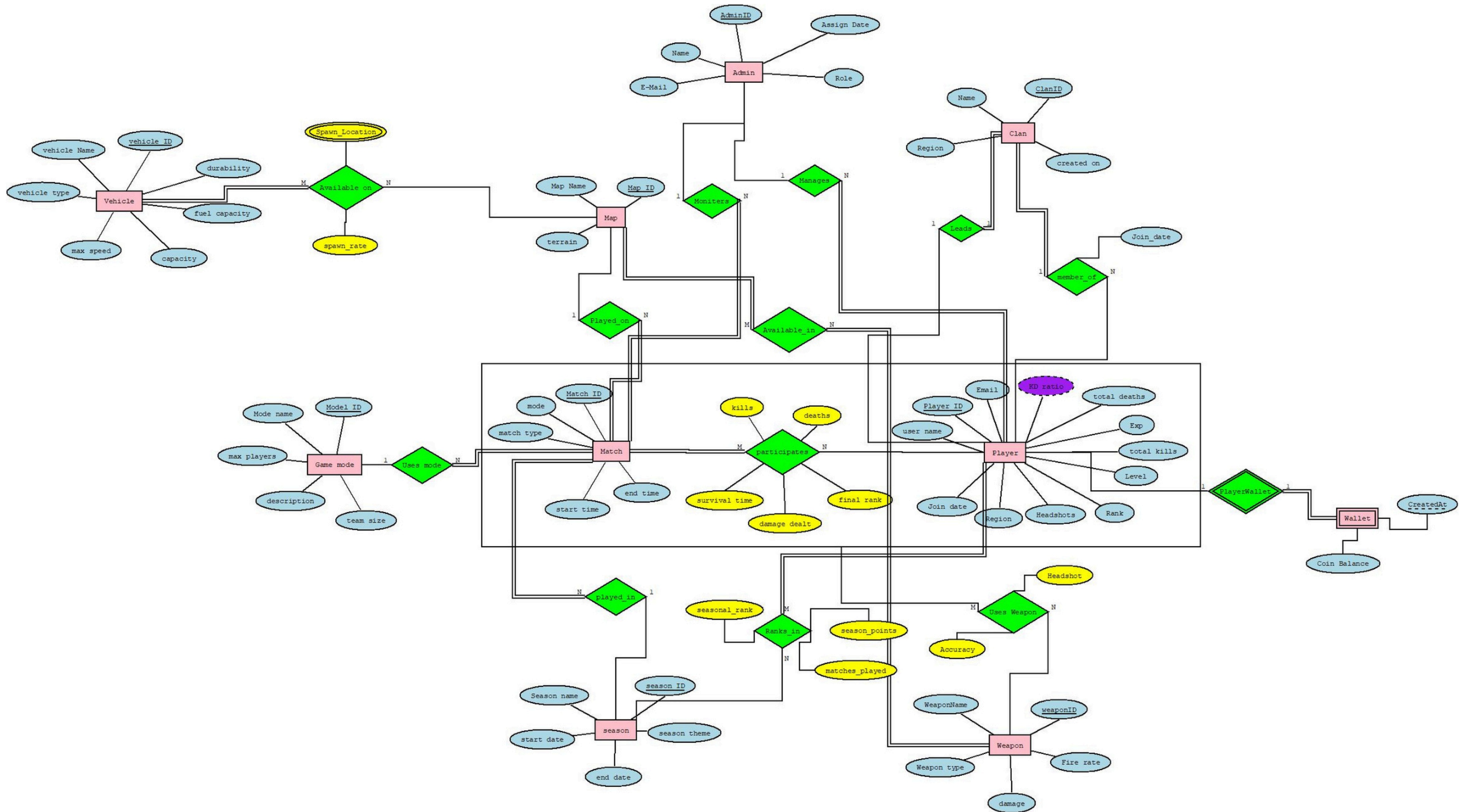
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PROJECT RELATIONAL SCHEMA

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MBA: MODERN BATTLE ARENA

INTRODUCTION

In the modern era of online multiplayer gaming, efficient data management plays a crucial role in tracking player performance, game statistics, and overall system functionality. This project focuses on the design and implementation of a relational database system for a battle royale-style game environment. The system is developed using PostgreSQL and is structured to handle various aspects of the game such as players, matches, weapons, vehicles, clans, and seasonal statistics.

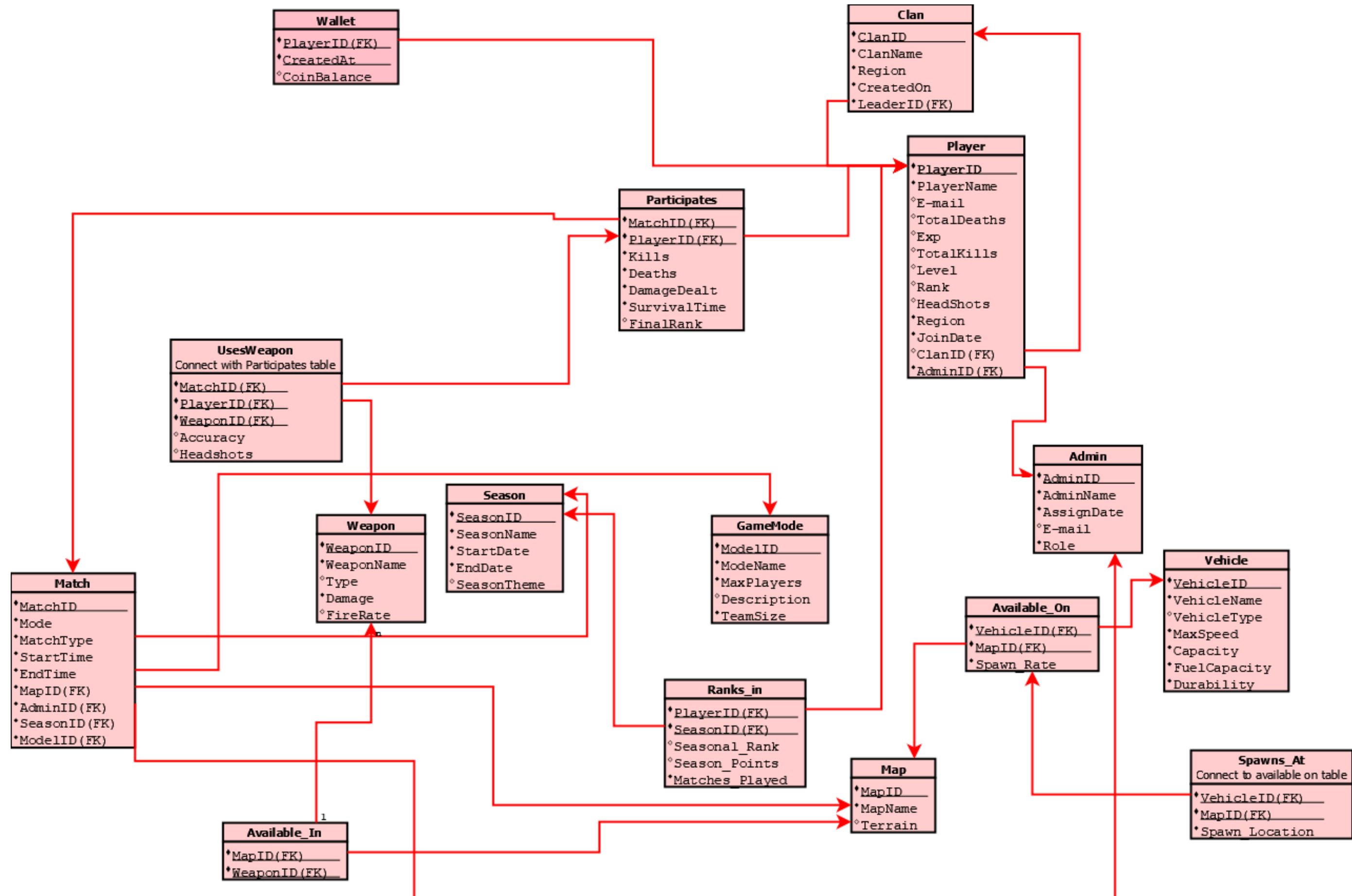
The primary objective of this project is to create a well-organized and normalized database schema that ensures data integrity, reduces redundancy, and supports efficient querying. The database captures detailed information about players, including their performance metrics such as kills, deaths, rank, and survival time. It also manages relationships between different entities, such as player participation in matches, clan memberships, and usage of weapons and vehicles.

Additionally, the system incorporates administrative control through admin entities and supports financial tracking through player wallets. The inclusion of seasonal data allows the system to maintain historical performance records across different time periods, making it scalable and adaptable for real-world gaming scenarios.

The database is designed following normalization principles, ensuring that it adheres to higher normal forms such as BCNF wherever applicable. Relationships such as one-to-many and many-to-many are carefully modeled using foreign keys and junction tables, providing a robust and flexible structure.

Overall, this project demonstrates the practical application of database design concepts in a real-world inspired scenario, highlighting the importance of structured data management in complex systems like online multiplayer games.

RELATIONAL SCHEMA



MINIMAL FD SET

Admin:	AdminID → AdminName, AssignDate, Email, Role
Clan:	ClanID → ClanName, Region, CreatedOn, LeaderID
Player:	PlayerID → PlayerName, Email, TotalDeaths, Exp, TotalKills, Level, Rank, HeadShots, Region, JoinDate, ClanID, AdminID
Wallet:	(PlayerID,CreatedAt) → CoinBalance
Map:	MapID → MapName, Terrain
Season:	SeasonID → SeasonName, StartDate, EndDate, SeasonTheme
GameMode:	ModelID → ModeName, MaxPlayers, Description, TeamSize
Weapon:	WeaponID → Name, Type, Damage, FireRate
Vehicle:	VehicleID → VehicleName, VehicleType, MaxSpeed, Capacity, FuelCapacity, Durability
Match:	MatchID → Mode, MatchType, StartTime, EndTime, MapID, AdminID, ModelID, SeasonID
Participates:	(MatchID, PlayerID) → Kills, Deaths, DamageDealt, SurvivalTime, FinalRank
UsesWeapon:	(MatchID, PlayerID, WeaponID) → Accuracy, Headshots
Ranks_In:	(PlayerID, SeasonID) → Seasonal_Rank, Season_Points, MatchesPlayed
Available_On:	(VehicleID, MapID) → SpawnRate

PROOF THAT RELATION IS BCNF

A relation is in Boyce–Codd Normal Form (BCNF) if and only if for every non-trivial functional dependency $X \rightarrow Y$, the determinant X is a superkey of the relation. BCNF is stricter than Third Normal Form and removes all update, insertion, and deletion anomalies that arise when a non-superkey attribute determines another attribute.

1.Admin:

FDs:

$\text{AdminID} \rightarrow \text{AdminName}, \text{AssignDate}, \text{Email}, \text{Role}$

Candidate Key: AdminID

Every FD's left side is a superkey $\rightarrow$ BCNF

2.Clan:

FDs:

$\text{ClanID} \rightarrow \text{ClanName}, \text{Region}, \text{CreatedOn}, \text{LeaderID}$

Candidate Key: ClanID

Every FD's left side is a superkey $\rightarrow$ BCNF

3.Player:

FDs:

$\text{PlayerID} \rightarrow \text{PlayerName}, \text{Email}, \text{TotalDeaths}, \text{Exp}, \text{TotalKills}, \text{Level}, \text{Rank}, \text{HeadShots}, \text{Region}, \text{JoinDate}, \text{ClanID}, \text{AdminID}$

Candidate Key: PlayerID

Every FD's left side is a superkey $\rightarrow$ BCNF

4.Wallet:

FDs:

(PlayerID, CreatedAt) → CoinBalance

Candidate Key: PlayerID

Every FD's left side is a superkey → BCNF

5.Map:

FDs:

MapID → MapName, Terrain

Candidate Key: MapID

Every FD's left side is a superkey → BCNF

6.Season:

FDs:

SeasonID → SeasonName, StartDate, EndDate, SeasonTheme

Candidate Key: SeasonID

Every FD's left side is a superkey → BCNF

7.GameMode:

FDs:

ModeID → ModeName, MaxPlayers, Description, TeamSize

Candidate Key: ModeID

Every FD's left side is a superkey → BCNF

8.Weapon:

FDs: WeaponID → Name, Type, Damage, FireRate

Candidate Key: WeaponID

Every FD's left side is a superkey → BCNF

9.Vehicle:

FDs: VehicleID → VehicleName, VehicleType, MaxSpeed, Capacity, FuelCapacity, Durability

Candidate Key: VehicleID

Every FD's left side is a superkey → BCNF

10.Match:

FDs: MatchID → Mode, MatchType, StartTime, EndTime, MapID, AdminID, ModelID, SeasonID

Candidate Key: MatchID

Every FD's left side is a superkey → BCNF

11.Participates:

FDs: MatchID, PlayerID → Kills, Deaths, DamageDealt, SurvivalTime, FinalRank

Candidate Key: (MatchID, PlayerID)

Every FD's left side is a superkey → BCNF

12.UsesWeapon:

FDs: MatchID, PlayerID, WeaponID → Accuracy, Headshots

Candidate Key: (MatchID, PlayerID, WeaponID)

Every FD's left side is a superkey → BCNF

13.Ranks_In:

FDs:

PlayerID, SeasonID $\rightarrow$ Seasonal_Rank, Season_Points, MatchesPlayed

Candidate Key: (PlayerID, SeasonID)

Every FD's left side is a superkey $\rightarrow$ BCNF

14.Available_In:

FDs:

MapID, WeaponID $\rightarrow$ (no non-key attributes)

Candidate Key: (MapID, WeaponID)

No non-trivial FDs exist $\rightarrow$ BCNF

15.Available_On:

FDs:

VehicleID, MapID $\rightarrow$ SpawnRate

Candidate Key: (VehicleID, MapID)

Every FD's left side is a superkey $\rightarrow$ BCNF

16.Spawns_At:

FDs:

VehicleID, MapID, SpawnLocation $\rightarrow$ (no non-key attributes)

Candidate Key: (VehicleID, MapID, SpawnLocation)

No non-trivial FDs exist $\rightarrow$ BCNF